
LAU
الجامعة اللبنانية الأميركية
Lebanese American University

Lecture 11

TDM

- Time Division Multiplexing (TDM): is the time interleaving of samples from several sources so that the information from these sources can be transmitted serially over a single communication channel.

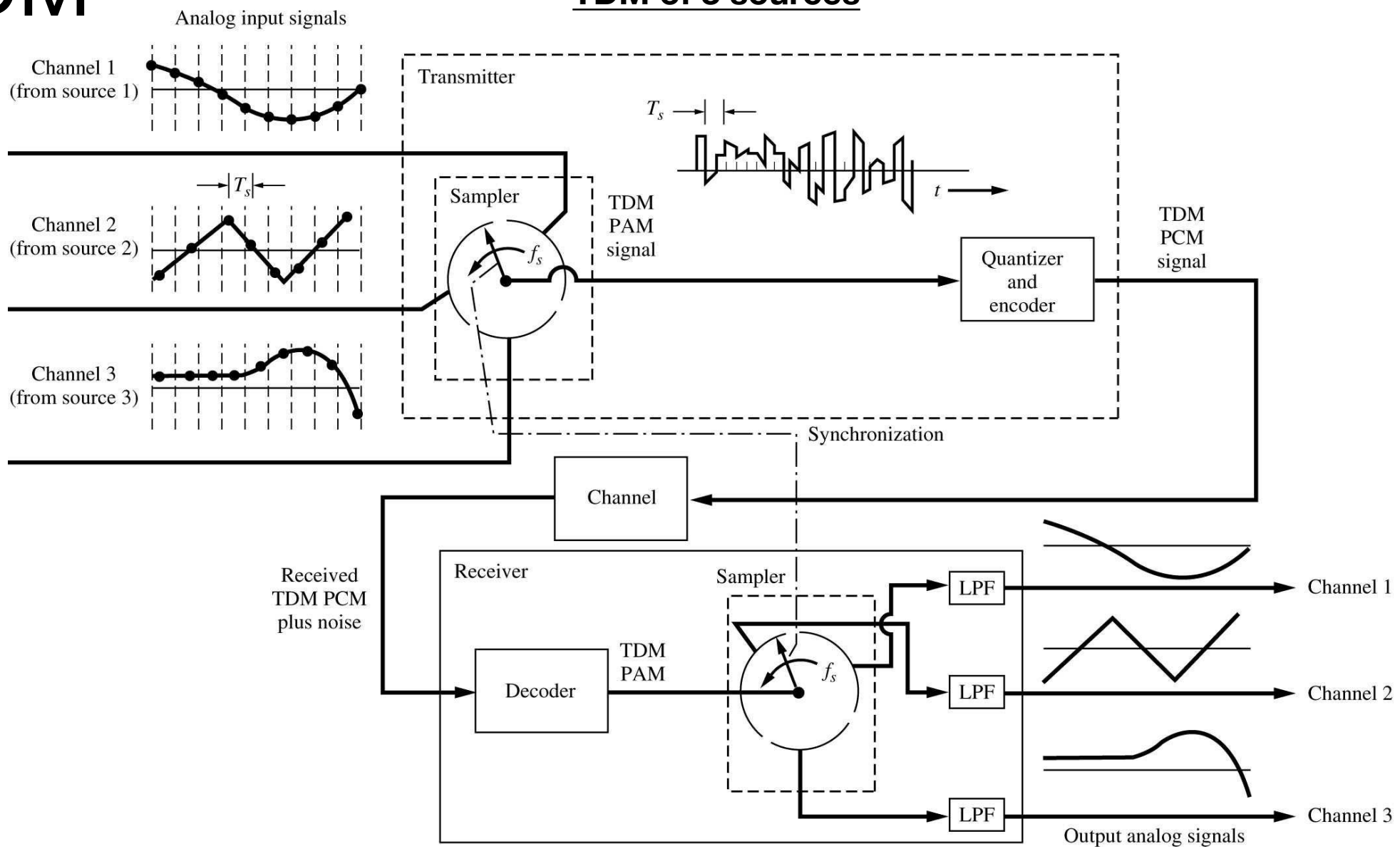
TDM

- Time Division Multiplexing (TDM) is a method where multiple signals share the same communication channel by allocating different time slots to each signal.
  - The channel is divided into small time intervals.
  - Each source sends its data only during its assigned time slot.
  - The receiver knows which slot belongs to which source and reconstructs the original signals.

- Example: Imagine three people sharing one microphone:
  - Person A speaks only during slot 1.
  - Person B speaks only during slot 2.
  - Person C speaks only during slot 3.

This repeats in cycles so that all three can share the same microphone without talking over each other.

TDM


TDM of 3 sources
Analog input signals
Channel 1
(from source 1)
Channel 2
(from source 2)
Channel 3
(from source 3)
T_s
Transmitter
T_s
t
Sampler
f_s
TDM
PAM
signal
Quantizer
and
encoder
TDM
PCM
signal
Synchronization
Channel
Received
TDM PCM
plus noise
Receiver
Sampler
LPF
LPF
LPF
Channel 1
Channel 2
Channel 3
Decoder
TDM
PAM
f_s
Output analog signals

TDM

- Consider the TDM of N_c sources.
- Denote by f_s the frequency of rotation of the commutator. f_s must satisfy the **Nyquist rate** for the analog source with the largest bandwidth:

$$f_s \geq 2 \max_{i=1 \ \ldots \ N_c} B_i$$

where B_i is the bandwidth of the *i*-th source.

- The pulse width of the TDM PAM signal is $\frac{T_s}{N_c}$.
- The pulse width of the TDM PCM signal is $\frac{T_s}{N_c n}$ where n is the number of bits in a PCM codeword.
- At the receiver, the decommutator must be synchronized with the incoming waveform so that samples from the *i*-th source will appear at the output of the i-th channel for $i = 1, \ldots, N_c$.
- ISI will cause PCM samples from one channel to appear at the output of another channel.

TDM

- In standard **TDM**, all sources are given **equal time slots**.
  - The sampling rate is determined by the **highest-bandwidth source** (Nyquist requirement).
- If one source has much higher bandwidth than others, this forces the commutator (switch) to run very fast — even for sources that don't need it.
- Solution: Instead of giving each source exactly one slot per frame:
  - Assign multiple slots per frame to high-bandwidth sources.
  - Assign fewer slots to low-bandwidth sources.
  - This is sometimes called asynchronous or weighted TDM.

TDM

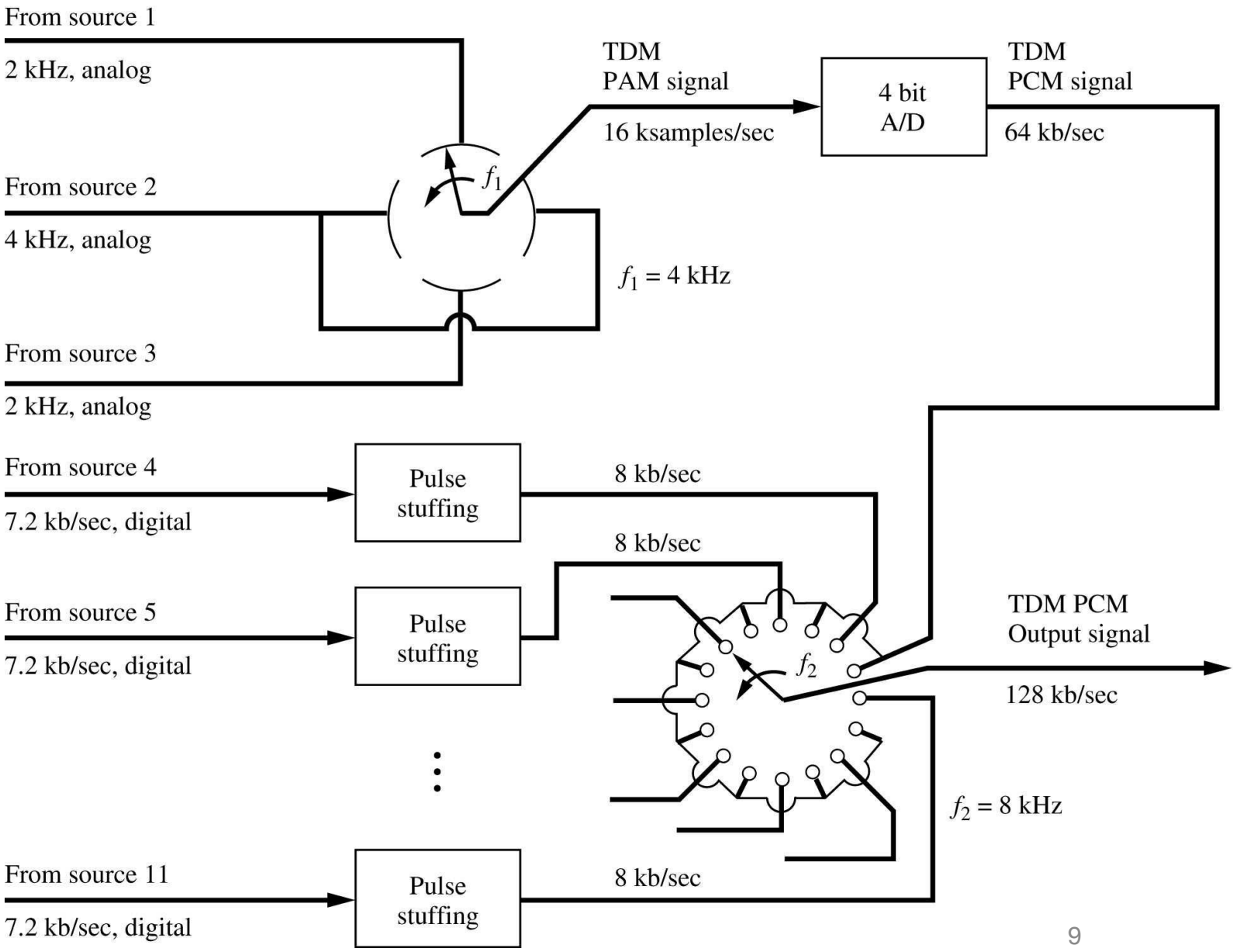
- Consider the case where the bandwidths of the different sources are markedly different.
- Instead of using a very fast commutator whose sampling rate is determined by the source having the largest bandwidth, the larger bandwidth sources may be connected to several switch positions of the sampler.

- In this case, large bandwidth sources are sampled more often than sources having smaller bandwidths.
  - Large-bandwidth sources are sampled more often (to capture their details without aliasing).
  - Small-bandwidth sources are sampled less often (enough to meet Nyquist, but no more).
- This relaxes the constraint on the sampling frequency f_s

TDM: Example

- Example on the design of a Time Division Multiplexer:
  - Design a TDM system that accommodates 11 sources having the following specifications:
    - Source 1: Analog, 2-kHz bandwidth.
    - Source 2: Analog, 4-kHz bandwidth.
    - Source 3: Analog, 2-kHz bandwidth.
    - Sources 4-11: Digital, synchronous at 7.2-kbits/s.
  - The design is given in the next figure. Note that the design is not unique.

TDM: Example

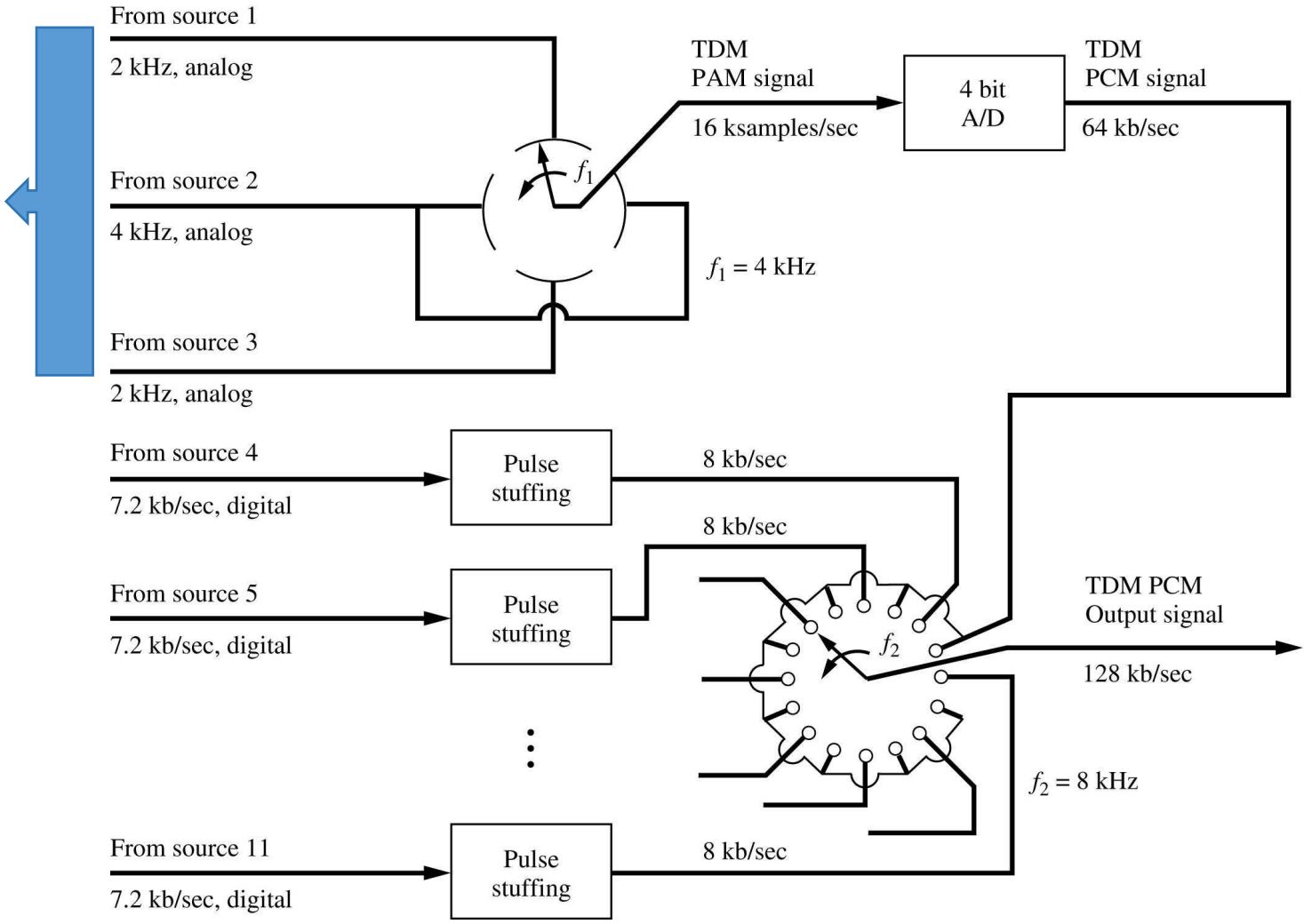

From source 1
2 kHz, analog
From source 2
4 kHz, analog
From source 3
2 kHz, analog
f_1
f_1 = 4 kHz
TDM
PAM signal
16 ksamples/sec
4 bit
A/D
TDM
PCM signal
64 kb/sec
From source 4
7.2 kb/sec, digital
From source 5
7.2 kb/sec, digital
From source 11
7.2 kb/sec, digital
Pulse
stuffing
Pulse
stuffing
Pulse
stuffing
8 kb/sec
8 kb/sec
8 kb/sec
f_2
f_2 = 8 kHz
TDM PCM
Output signal
128 kb/sec

TDM: Example solution

1. Sampling requirement (Nyquist)

•Source 1: Bandwidth 2 kHz--> fs≥4 kHz
•Source 2: Bandwidth 4 kHz4 → fs≥8 kHz
•Source 3: Bandwidth 2 kHz→ fs≥4 kHz
Instead of rotating the commutator at 8 kHz for everyone, they:

- •Rotate at f1=4 kHz
- •Sample Source 2 twice per rotation (so it effectively gets 8 kHz).



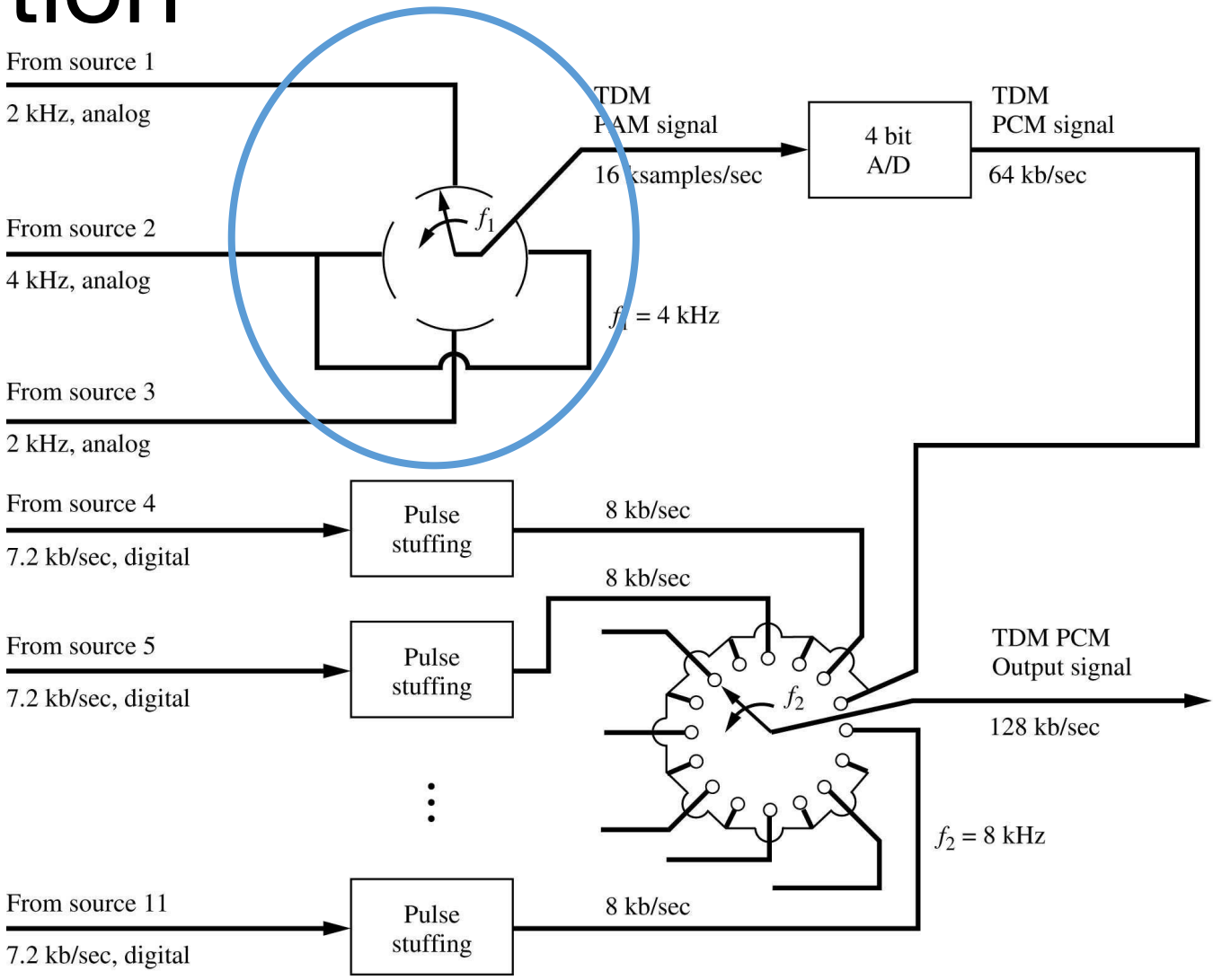
TDM: Example solution

2. First commutator output (TDM-PAM signal)

- In one rotation, the commutator connects to:
  - Source 1 once
  - Source 2 twice
  - Source 3 once

→ **Total = 4 samples per rotation**

So, total sample rate = 4f1=4×4 kHz=16 k samples/s



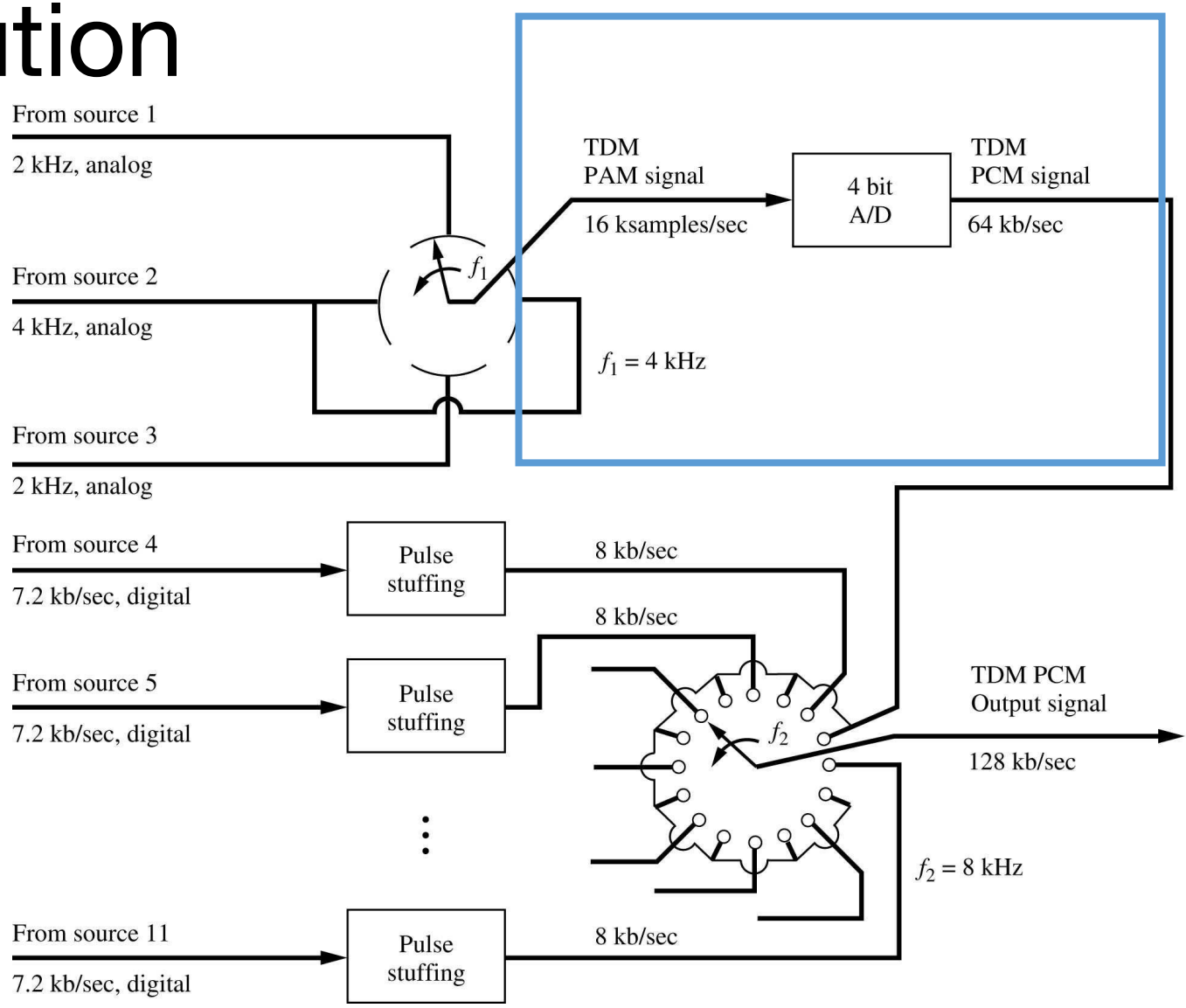
TDM: Example solution

3. Pulse width for TDM-PAM

•If the commutator rotates at 4 kHz and there are 4 samples per rotation:
Pulse width=1/4f1=1/16 kHz This is the width of each PAM pulse in the TDM signal.

4. Converting to digital (PCM)

•Each sample is quantized into n=4 bits (PCM codeword).
•Data rate after quantization:
- •Data rate=(16 k samples/s)×(4 bits/sampl e)=64 kbits/s



TDM: Example solution

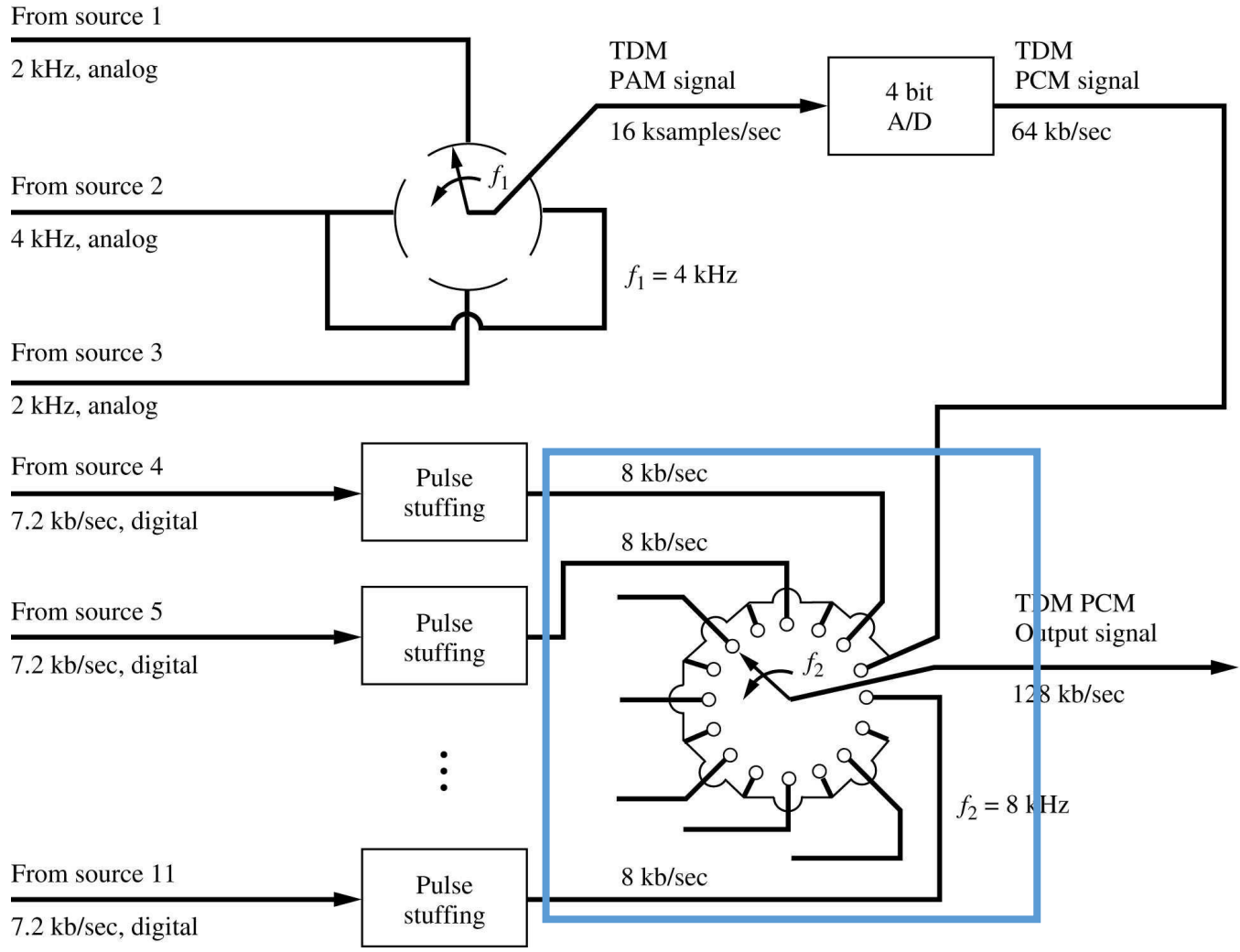
- To satisfy the Nyquist rate, the analog sources must be sampled at 4 kHz, 8 kHz and 4 kHz respectively.
  - This can be accomplished by rotating the first commutator at $f_1 = 4$ kHz and by sampling source 2 twice per rotation.

- The pulse width of the TDM PAM signal at the output of the first commutator is equal to $1/4f1$.
  - This produces a $4f_1 = 16$ ksamples/s TDM PAM signal on the output of the first commutator.

- Assume that each sample is converted to a n = 4 bit PCM codeword.
  - The pulse width of the TDM PCM signal is equal to $1/4f1n$ resulting in a data rate of $4f_1n = 64$ kbits/s at the output of the A/D converter.

TDM: Example solution

The digital data at the output of the A/D converter may be merged with the data from the digital sources using a second commutator rotating at f_2 = 8 kHz.

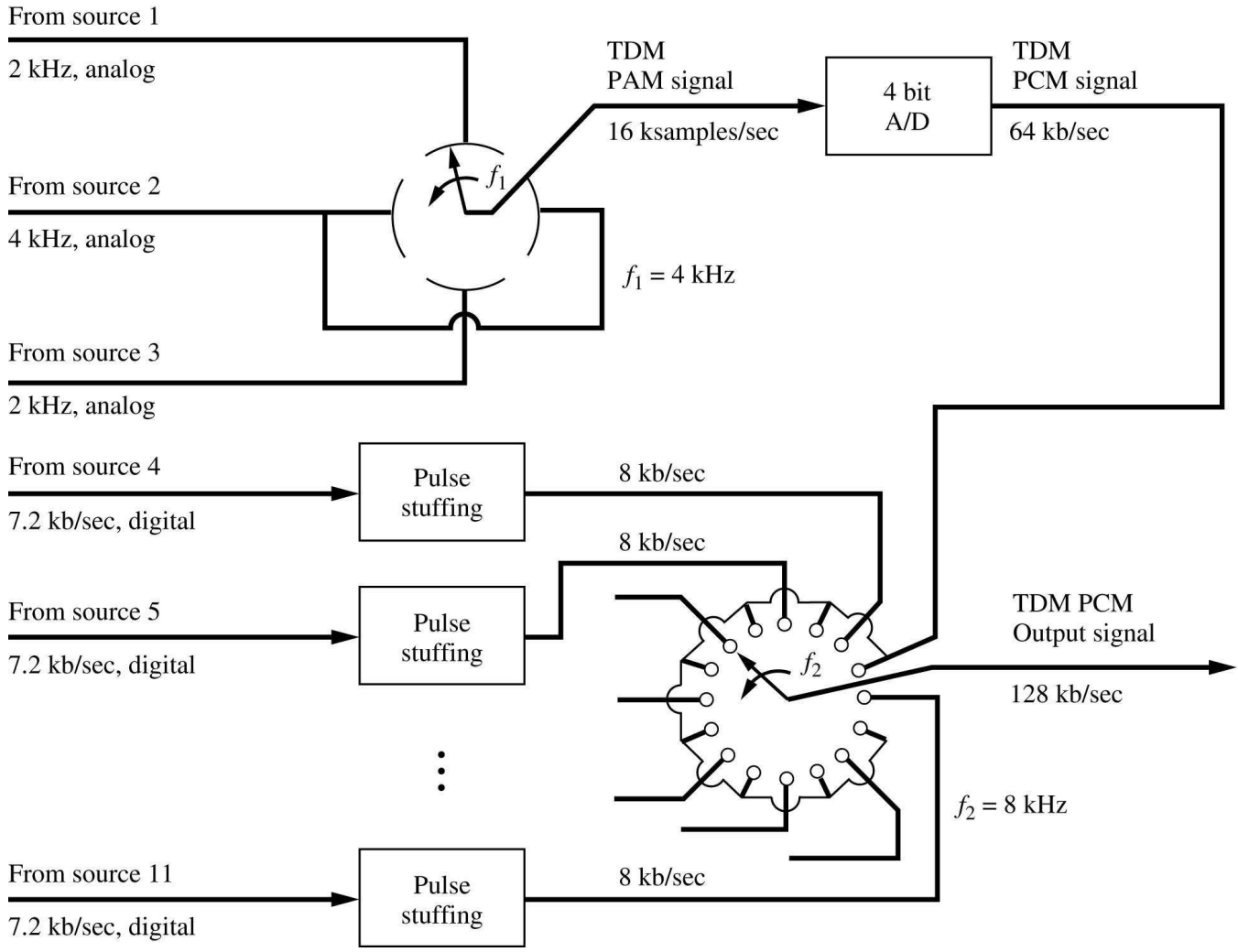


TDM: Example solution

1.The 64 kbps stream from the first stage (PCM of the 3 analog sources).

- In the second commutator, each terminal (slot) outputs 8,000 bits per second because: Slot rate=f2=8 kHz.
- This is fixed for all slots because the second commutator rotates at f2=8 kHz

- Analog PCM stream: 64 kbps ÷ 8 kbps per slot = 8 slots needed.



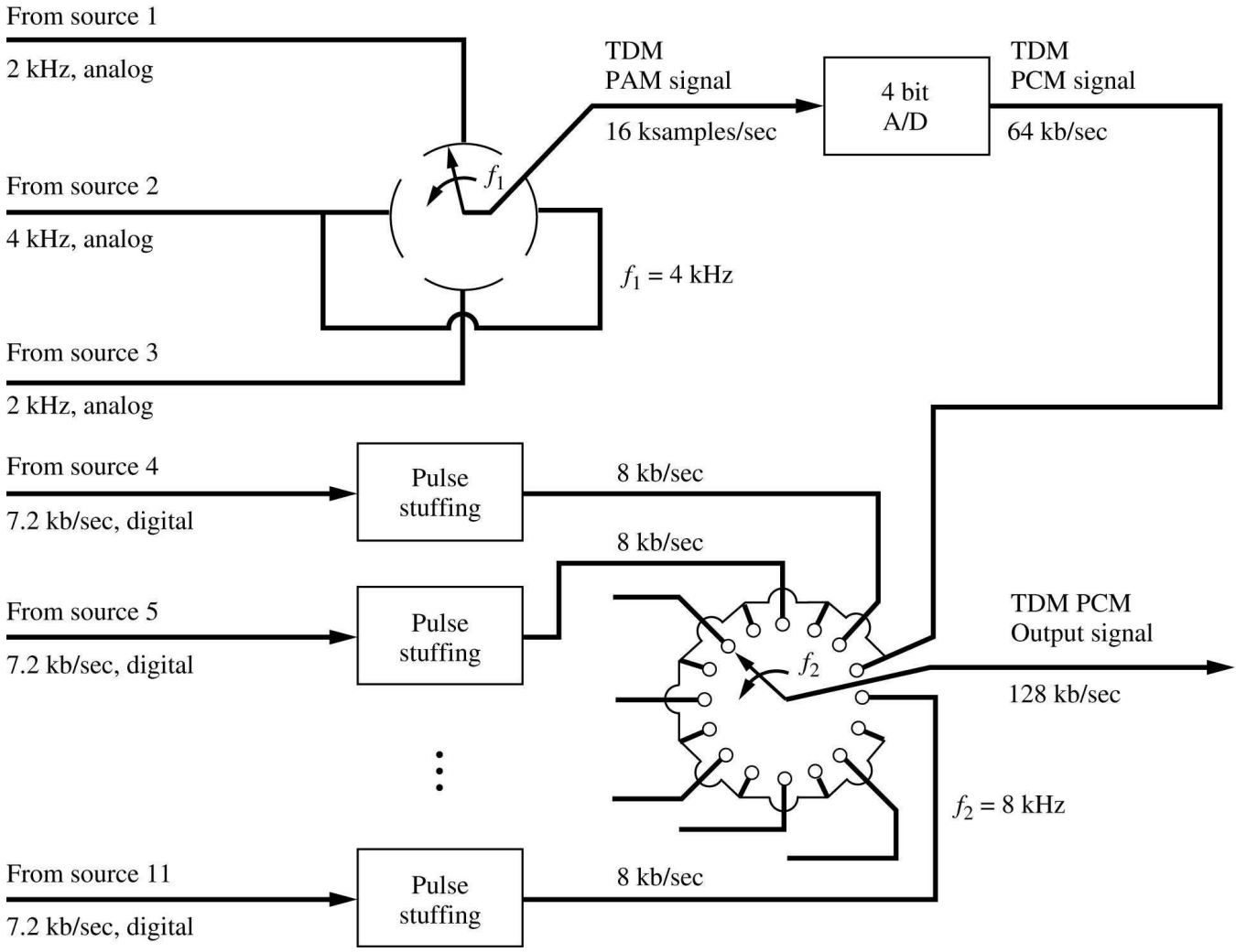
TDM: Example solution

2. The digital sources (Sources 4–11), each at 8 kbps after pulse stuffing.

•Digital sources: 8 sources × 1 slot each = 8 slots needed.

Total slots=8 (analog)+8 (digital)=16

→ 16 terminals for the second commutator.

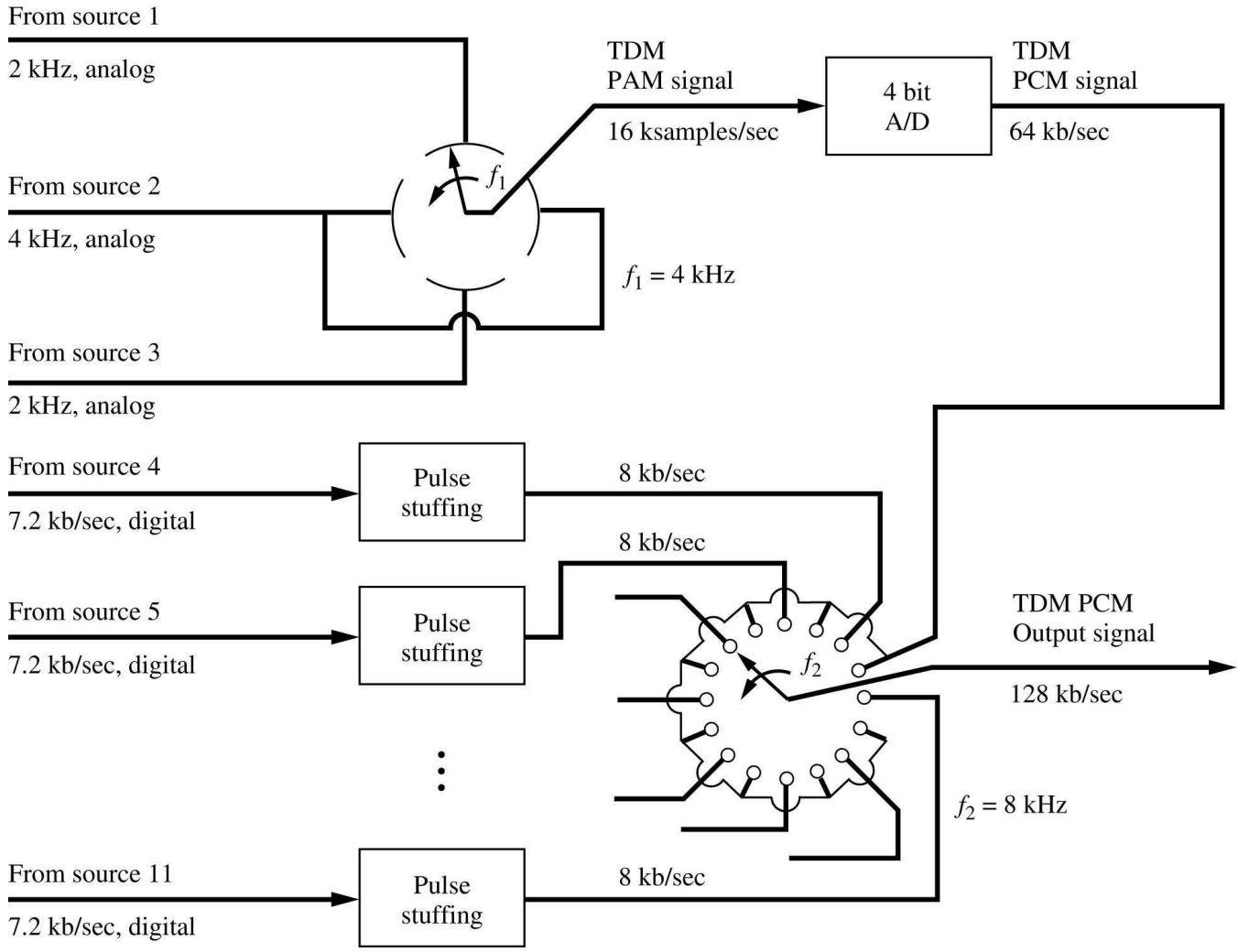


TDM: Example solution

The 64 kbits/s signal is connected to 8 (out of the 16) terminals of the commutator.

Each one of the sources 4-11 is connected to one terminal of the second commutator.

Since the digital sources are supplying 7.2 kbits/s data streams, pulse stuffing is used to raise the rate of each source to 8 kbits/s.



TDM: Example solution

- Pulse stuffing is a method used in Time Division Multiplexing (TDM) to make sure all channels fit into a synchronous TDM frame, even when their actual bit rates are slightly different.
  - In synchronous TDM, every channel is given a fixed time slot in each frame.
  - The multiplexer assumes each channel sends the same number of bits per second.
  - If one source sends slightly less data than the slot allows, the frame timing will go out of sync.

- Example:
  - Digital source rate: 7.2 kbps
  - Slot capacity: 8 kbps
  - Difference: 8.0−7.2=0.8 kbps
  - We insert 0.8 kbps worth of extra bits into each source's data stream so it perfectly fills the 8 kbps slot.

TDM: Example solution

The bit rate at the output of the multiplexer is 16 × 8 = 128 kbits/s.

1. Frame structure

- 16 terminals (slots) per frame.
- Each slot carries 1 bit every time the commutator rotates.
- Commutes (rotates) at f2 =8 kHz → visits each slot 8,000 times per second.

2. Slot bit rate

- Each slot = 8,000 visits/sec×1 bit/visit=8 kbits/s.

3. Total bit rate

- 16 slots × 8 kbps per slot: Total bit rate=16×8 kbits/s=128 kbits/s.

Frame Synchronization

- In TDM, the receiver must know where each frame starts to correctly separate the channels.
- Without synchronization, the receiver might start reading in the wrong position → data from different channels gets mixed.
- Frame synchronization can be provided to the receiver demultiplexer either by:
  - Transmitting a frame sync signal over a separate channel. This solution is trivial and not economical.
  - Providing frame sync in the TDM signal itself. In what follows,we consider this solution.

Frame Synchronization

Two ways to provide frame synchronization

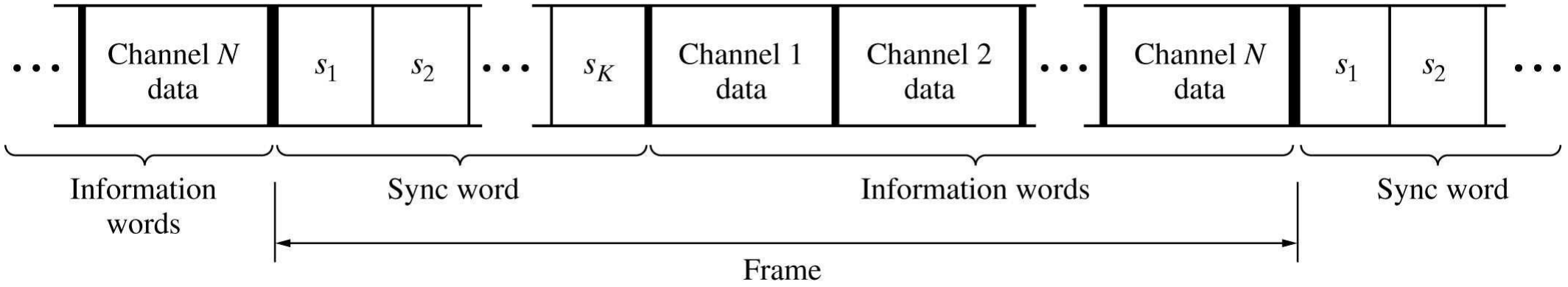
- **Separate channel method**
  - Send a dedicated synchronization signal over another line.
  - Easy to do but **wastes bandwidth** → not efficient.

- **Embed sync into the TDM signal (preferred method)**
  - Add a **special bit pattern** inside the TDM stream.
  - Receiver detects this pattern to know when a new frame starts.

Frame Synchronization

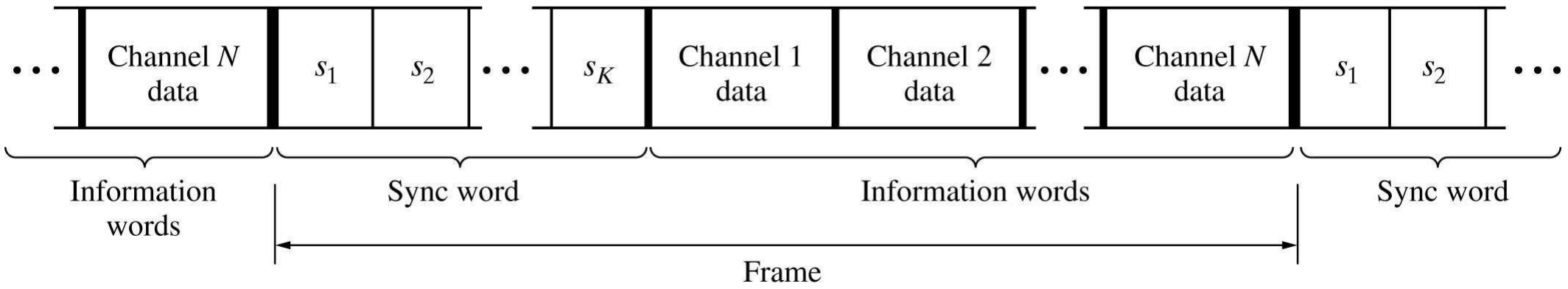
Frame sync may be multiplexed along with the information words. A unique K bit sync word is transmitted at the beginning of each frame that contains the information from N sources.

- At the start of each frame, insert a K-bit unique pattern (sync word).
- This pattern is known to both transmitter and receiver.
- The receiver looks for this pattern in the incoming bit stream:
  - When it finds it → frame start detected.
  - Then it reads N channel data blocks in order until the next sync word.



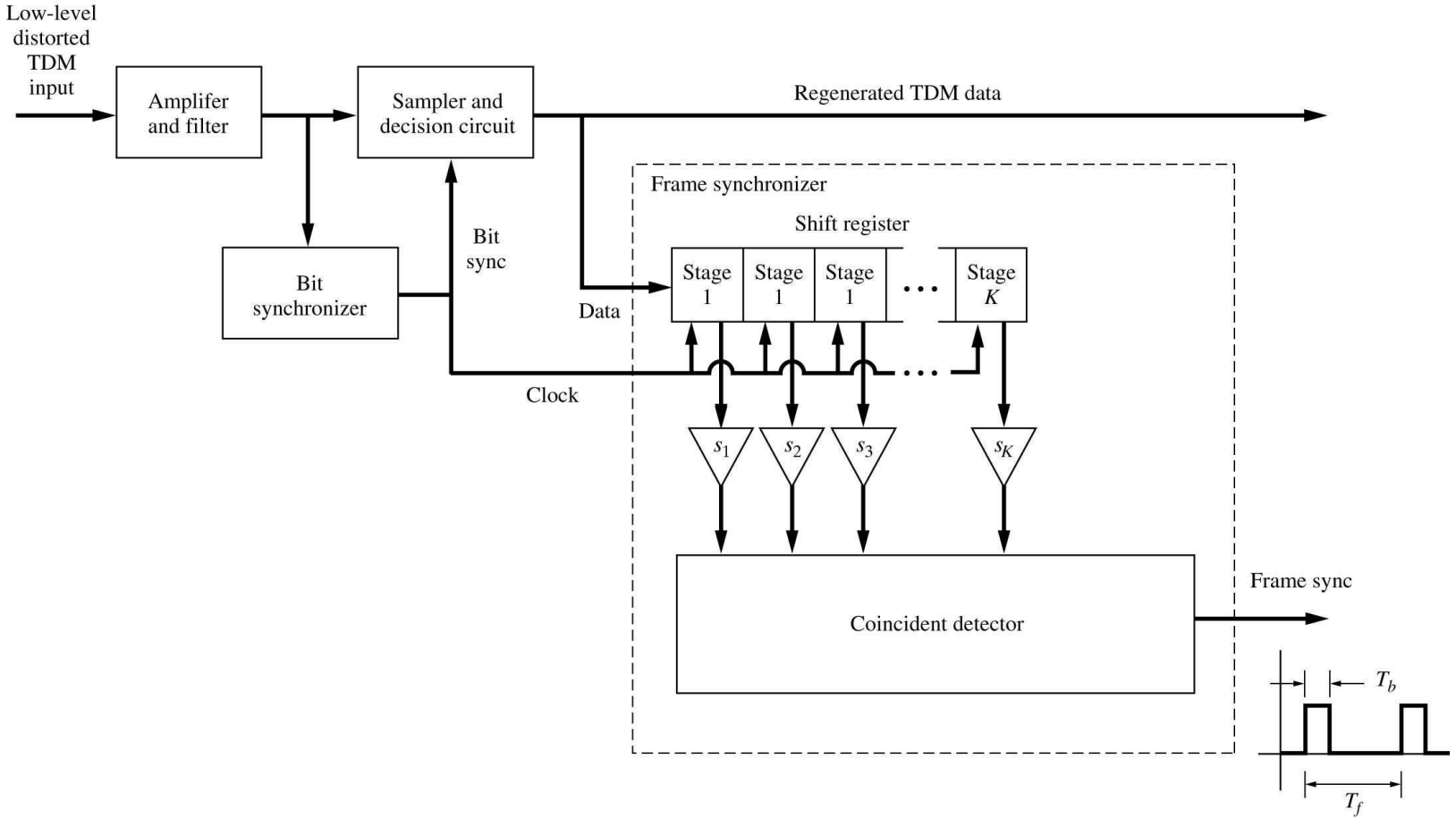
Frame Synchronization

- Information words = actual data from each channel in the frame.
- Sync word = fixed sequence of K bits (s1,s2,...,sK) marking the start of a frame.
- Pattern repeats at the start of each frame.



Frame Synchronization

- At the receiver side, a frame synchronizer circuit is used. It cross-correlates the TDM signal with the expected unique sync word s = (s1, . . . ,sK).



Frame Synchronization

- The sj ’s within the triangles below the shift registers denote the presence or absence of an inverter:
  - If sj = 0 then there is an inverter. Otherwise, there is no inverter.
- The coincident detector is a K-input AND gate.
- The frame sync signal is recovered because:
  - If the unique sync word is present in the shift register, all the inputs of the coincident detector will be equal to 1. Consequently, the output of the detector is equal to 1. This binary 1 will result in a high voltage level for a duration of Tb seconds.
  - If the sync word is not present in the shift register, at least one input of the coincident detector is equal to 0 resulting in a zero-level output.

Frame Synchronization

A false sync occurs if K successive information bits happen to be equal to the sync word. Forequiprobable TDM data, this an happen with probability:

$$P_f = \left(\frac{1}{2}\right)^K = 2^{-K}$$

In frame sync design, this equation is used to determine the length K of the sync word so that the false sync probability will not exceed a certain tolerated value.
For example, if Pf = 4 × 10–5 is the allowed probability of false sync, then a K = 15-bit sync word is required.